$$\int_{-T}^{T} \cos\left(\frac{\pi n}{T} x\right) dx = \begin{cases} x & \text{for } \frac{\pi n}{T} = 0 \\ \frac{T}{\pi n} \sin\left(\frac{\pi n}{T} x\right) & \text{otherwise} \end{cases}$$

$$\int_{-2}^{2} \frac{\log x}{x-1} \, \mathrm{d}x = -\operatorname{Li}_2\left(e^{i\pi}\right) + \operatorname{Li}_2\left(-3e^{i\pi}\right)$$

$$\lim_{x\to\infty} \frac{x^2}{2x^2+2} = \frac{1}{2}$$

$$\lim_{x\to 0^+} \left(\frac{1}{x} \sin(x)\right) = 1$$

$$\sum_{k=1}^{\infty} \frac{k}{k+1} = \infty$$

$$\oiiint_V \mathrm{d}v$$

$$\int_{-1}^{1} \log x \, \mathrm{d}x = x \log(x) - x$$

$$\int_{-1}^{1} \log x \, \mathrm{d}x = x \log x - x$$

$$\Delta = -3^2 - 1 \cdot$$

$$\left(\begin{array}{cc|c} 2 & 1 & 3 \\ -1 & 7 & 2 \end{array}\right)$$

$$\lim_{x\to -1} \frac{x}{x^2-1}$$

$$\int_{-1}^{1} x^2 \, \mathrm{d}x = \left.\frac{x^3}{3}\right|_{-1}^{1} = \frac{2}{3}$$

$$\int_{-1}^{1} \log x \, \mathrm{d}x = x \log x - x|_{-1}^{1} = 2 - \pi i$$

$$\lim_{x\to\infty} \left(x^3 - 3x^2 + 5x - 12\right) = \lim_{x\to\infty} 3x^3 = \infty$$

$$\int_{-\infty}^{\infty} \frac{1}{x^4+1} \, \mathrm{d}x = \frac{\sqrt{2}\pi}{2}$$

$$\sum_{k=1}^{\infty} \binom{n+1}{n} x^n = \frac{1}{x^n}$$

$$\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$$

$$\int_{-1}^{1} \frac{x}{x^2+1}\,\mathrm{d}x = \frac{1}{2}\log\left(x^2+1\right)\Big|_{-1}^{1} = \log 2 = 0$$

$$x^2 - 3x + 2 = 0$$

$$7x - 8 = 0 \int_{-\infty}^{\infty} \frac{x}{e^x}\,\mathrm{d}x$$

$$\sum_{k=-\infty}^{\infty} \frac{(-1)^k}{k^4+1}$$

$$\int_{-x}^{x} t - t^2\,\mathrm{d}t = \frac{2x^3}{3} =$$

$$\int_{-\infty}^{\infty} \frac{1}{x^4+1}\,\mathrm{d}x$$

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + \left(x^2-\alpha^2\right)y = 0$$

$$J_\alpha(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+\alpha+1)}\left(\frac{x}{2}\right)^{2m+\alpha}$$

$$\frac{\sin(x)}{x} = \prod_{n=1}^{\infty} \cos\left(\frac{x}{2^n}\right)$$